

EVANGEL IHEUKWUMERE

Software & Systems Engineer

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Location: United Kingdom | Open to: Remote, Hybrid, On-site

PROFESSIONAL SUMMARY

Final-year Software Engineering student (BSc Hons, University of Portsmouth, predicted First, 2026) with commercial experience delivering production-grade systems across full-stack development, AI/ML integration, and data-intensive applications. Industrial placement at Eli Lilly and Company, where contributions to enterprise chatbot systems supported an estimated \$4M+ in operational savings. Hands-on experience building and training AI models using Amazon Bedrock, creating Lambda function agents, and integrating LLM inference endpoints via Cortex in a regulated pharmaceutical environment. Currently building data visualisation and call analytics tooling at Marchex Inc., processing millions of records for enterprise clients including General Motors. Co-founded and sole-built Kevvlar, a SaaS platform with ~150 users, 200+ workspaces, real-time collaboration, and payment infrastructure. Platform remains live after approximately 5 years. Currently developing Gloomhunt, a browser-based multiplayer game with real monetary transactions, built solo from scratch including a custom game engine, deterministic anti-cheat system, double-entry financial ledger, and an 8-page admin console (~50,000+ lines of code, 230+ automated tests). Additional experience in hardware prototyping, including UAV design and inertial stabilisation systems for model rocketry.

This is a Core CV intended as a master reference document. It is not tailored to a specific role.

EDUCATION

BSc (Hons) Software Engineering

Expected 2026

University of Portsmouth, United Kingdom

- Predicted First. Final-year modules include Advanced Database Concepts, Complex Problem Solving, Software Engineering Culture, and Project Management.
- Industrial placement year completed at Eli Lilly and Company (July 2024 - July 2025).

West African Examination Council (WAEC)

2018

Heritage Secondary School, GCSE Equivalent

- Computer Science (A), Maths (B), Chemistry (B), Physics (B).

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer (Remote)

June 2025 - Present

Marchex Inc.

- Developing data-intensive web applications using Angular and C#/.NET, processing millions of call analytics records for enterprise clients including General Motors across multi-branch operations.
- Built a call playback and transcript viewer enabling business users to review recorded conversations in a real-time chat format with synchronised audio playback.
- Collaborated with a senior engineer to design and build a flexible data visualisation engine allowing users to structure call analytics data dynamically and generate custom chart types via AG Grid.
- Collaborated on deploying and maintaining a new backend service for processing audio call data and delivering processed results to the client application.
- Shipped 35+ features and 15 bug fixes to production within sprint-based delivery cycles, working within a large-scale production codebase with established CI/CD workflows and coding standards.

Stack: Angular, TypeScript, C# / .NET, AG Grid, charting libraries, Git

Software Engineer, Industrial Placement

July 2024 - July 2025

Eli Lilly and Company

- Integrated AI and ML capabilities into enterprise chatbot systems using Azure AI Language Studio, refining NLU training datasets and achieving 95% intent classification accuracy, exceeding the 92% production-readiness threshold.
- Contributed to chatbot systems that saved Eli Lilly an estimated \$4,000,000+ in out-of-hours customer support costs, as reported by management.
- Developed a React-based UAT platform that reduced chatbot testing time by over one full working day per cycle compared to the previous manual process. Tool was adopted team-wide and entered beta rollout with a multinational department for broader adoption.
- Integrated enterprise LLM endpoints via Cortex (Lilly's internal AI gateway), enabling dynamic response generation within chatbot dialogue flows in a regulated pharmaceutical environment.
- Built and trained an AI model using Amazon Bedrock to answer queries on painkiller active ingredients, surfacing information on drug compositions, active molecules, clinical trial data, and research findings from a custom knowledge base. The model was also capable of rendering drug molecular structures. Created AWS Lambda function agents to handle specific tasks for the AI. Presented the project in a lunch and learn session to fellow developers.
- Built a lightweight chatbot framework from scratch for an internal innovation challenge, demonstrating multi-turn conversation handling with configurable intent routing. Team placed runner-up.

- Earned SAFe Agile Practitioner certification. Gained hands-on experience with code reviews, CI/CD pipelines, and Agile/SAFe ceremonies within a regulated enterprise environment.

Stack: *React, JavaScript, Python, Azure AI Language Studio, Amazon Bedrock, AWS Lambda, Cortex (Lilly internal LLM gateway), Azure, AWS CloudFormation, CI/CD, SAFe Agile*

IT Help Adviser

September 2023 - June 2024

University of Portsmouth

- Provided first-line IT support to students and staff, troubleshooting issues with university systems, network connectivity, software installations, and account access.
- Assisted during onboarding periods, guiding new students through system setup, email configuration, and VPN access.
- Documented common issues and contributed to internal knowledge base articles to reduce repeat queries.

Intern Computing Teacher

June 2023 - July 2023

Bay House School & Sixth Form

- Assisted computing teachers in delivering lessons across Key Stage 3 and Key Stage 4 curricula, covering topics including programming fundamentals, data representation, and networking.
- Provided one-on-one support to students during practical coding sessions, helping debug code and explaining core concepts. Contributed to improved student engagement and learning outcomes.
- Obtained UK GDPR certification and UK Safeguarding Training Certificate (TSAT) as part of the role.

Contract Game Developer (Remote)

March 2023 - August 2023

SgtWoofy

- Authored developer-facing documentation for a PHP-based API service, including endpoint descriptions, request/response schemas, authentication flows, and usage examples.
- Engineered an in-game store system enabling users to purchase virtual assets with real currency, implementing purchase flow logic, inventory management, and payment processing API integration.
- Worked within an existing game engine/server architecture, adapting to the project's PHP backend and custom data layer.

Stack: *PHP, game engine scripting, API design, payment integration*

Full-Stack Engineer (Remote)

June 2022 - September 2022

Weekend Social

- Redesigned application UI based on feedback from the product design team, implementing responsive layouts and improving interaction patterns to increase usability.
- Developed an admin dashboard enabling company administrators to monitor user activity, review engagement metrics, and manage content. Received positive user feedback on usability improvements.
- Worked within a startup environment with short iteration cycles, shipping features directly to production.

Stack: *JavaScript, frontend frameworks, REST APIs*

CTO & Software Engineer (Co-founder)

June 2021 - February 2023

Kevvlar

- Co-founded and led technical development of Kevvlar, an all-in-one project management SaaS platform alongside the CEO/designer. Sole developer, building the entire application from Figma designs while simultaneously handling product roadmap, go-to-market strategy, and user feedback.
- Grew the platform to ~150 users and over 200 workspaces with 10 active subscribers. Launched on Product Hunt. Platform remains live and accessible.
- Architected the full-stack application: React frontend with Redux for global state management, Node.js/Express backend, MongoDB for data persistence, and Redis for caching and session management.
- Integrated Twilio APIs for real-time video conferencing and SMS notifications. Implemented WebSocket-based live chat with connection lifecycle management, reconnection logic, and message ordering.
- Implemented Stripe payment processing including subscription management, payment intent flows, and webhook handling for billing event processing.
- Built responsive, mobile-first UI with accessibility considerations. Implemented error handling and logging across the full stack.
- Managed the product roadmap, made architectural decisions, and coordinated with co-founders on feature prioritisation and user feedback loops.
- Handled deployment, infrastructure setup, and monitoring on AWS and Firebase.

Stack: *React, Redux, Node.js, Express, MongoDB, Redis, WebSockets, Twilio, Stripe, AWS, Firebase*

PROJECTS: HARDWARE, AEROSPACE & CONTROL SYSTEMS

Project Sentinel — Autonomous Fixed-Wing UAV

- Designed and partially built an autonomous fixed-wing UAV intended for security surveillance and area deterrence applications.
- Project scope included airframe selection and layout, sensor payload planning (camera, motion detection), and autonomous flight path logic.

- Used CAD tools (Fusion 360, SolidWorks) for airframe and component layout modelling.
- Researched flight controller integration, telemetry data handling, and autonomous navigation constraints (GPS waypoint following, return-to-home logic).
- Status: Prototype/design-stage. Not a completed production system.

Stack: *Fusion 360, SolidWorks, flight controller research, sensor integration planning, autonomous navigation concepts*

6-Axis Gyro Stabilisation Module — *Rocket Flight Correction*

- Designed and built a 6-axis IMU-based stabilisation module for real-time attitude correction in model rockets, computing corrective outputs to counteract roll, pitch, and yaw deviations during flight.
- Intended as an attachable subsystem to a rocket's main flight computer, feeding correction signals to actuators (e.g., thrust vectoring or fin servos).
- Involved embedded programming, sensor fusion concepts (complementary or Kalman filtering), and real-time data processing under tight timing constraints.
- Status: Functional prototype-stage module. Tested on bench; limited flight testing.

Stack: *IMU sensors (6-axis), embedded programming, sensor fusion, real-time control loops, PID tuning concepts, CAD (Fusion 360)*

PROJECTS: SOFTWARE

Gloomhunt — *Browser-Based Multiplayer Game Platform*

Live at: <https://gloomhunt.onrender.com/>

- Designing and building a full-stack browser-based top-down survival shooter where players compete in admin-defined challenges (bounties) to win real monetary prizes. Built entirely solo from scratch with no game engine, no client-side frameworks, and no dependencies. ~50,000+ lines of application code.
- Built a custom game engine in vanilla JavaScript and HTML5 Canvas: fixed 60Hz timestep with accumulator pattern, spatial hash collision detection, object pooling for zero GC pressure, seeded deterministic PRNG (mulberry32), camera system with screen shake, positional audio engine, and viewport scaling system supporting desktop and mobile.
- Implemented a complete combat system with 12 weapon types (7 fire patterns, 6 projectile behaviours), 5 gadget types with lob arc physics, 3 AI companion types, a critical hit system, combo/kill-streak system with milestone rewards, and a dash ability with i-frames and ghost trail.
- Built an enemy AI system with 12 enemy types across 4 archetypes, 6 AI movement styles (direct chase, flank, intercept, orbit, maintain distance, stalk), 7 enemy abilities (charge, ranged attack, cloak, split on death, heal, shield, summon), melee telegraph system, and steering-based obstacle avoidance via spatial hash queries.
- Developed a procedural world system with chunk-based seeded generation, destructible props (exploding barrels with chain reactions, loot crates), 15+ pickup types with magnet pull physics, and an SVG-to-canvas icon pipeline pre-rendering 39 icons at load time for zero per-frame cost.
- Built the server platform with Node.js, Express, and PostgreSQL (Supabase): 46+ database migrations (auto-run on start), ~50 RESTful API endpoints, Google OAuth2 authentication via Passport.js, Postgres-backed sessions, auth/admin middleware guards, CSRF protection, and centralised error handling.
- Implemented a double-entry financial ledger: append-only ledger entries as source of truth, system accounts (house revenue, bonus pool, reward escrow), hold/capture/release/refund lifecycle with idempotency keys, auto-payout pipeline for verified wins, and FULL OUTER JOIN reconciliation of wallet cache vs ledger-derived totals.
- Built a deterministic anti-cheat system: all gameplay uses seeded PRNG (never Math.random), so every game run is a pure function of seed + config + ordered inputs. Server re-simulates and compares end-state hashes. HMAC-signed checkpoints, gzip-compressed input logs, schema-versioned hashes, and MATCH/MISMATCH/ABORTED verdict system with batch re-verification.
- Developed a bounty system with revision-based lifecycle (immutable revisions, atomic publish/retire), 8 configurable win conditions per bounty, frozen config artifacts with canonical JSON hashing, 5-gate session start chain, and a settlement review queue with per-axis state guards and audit logging.
- Built an 8-page admin console SPA: financial dashboard with live metrics, bounty CRUD with 4-section editor and settlement queue, 456-field config editor across 10 lazy-rendered tabs, player management with wallet/sessions/ledger tabs, grants system with audience targeting, anti-cheat verification dashboard, and admin user management with safety guards.
- Implemented a meta-progression system: XP formula, 20 rank tiers (Hatchling to Apex), 29 achievements with session-based and lifetime milestones, shareable canvas-generated result cards (1080x1350) via Web Share API, and a 7-step guided tutorial system.
- Built responsive HUD system with auto-scaling, segmented ammo bars, 39 SVG icons with CSS mask tinting, left-hand mirror layout for mobile, dual-zone virtual joysticks with configurable deadzone, and near-miss feedback for failed bounty conditions.
- Wrote 230+ automated tests using Vitest covering admin CRUD, settlement pipeline, wallet operations, bounty conditions, config validation, anti-cheat verification, and client UI logic. Integration tests run against real Postgres with idempotency and concurrency verification.

Stack: *Vanilla JavaScript, HTML5 Canvas, Node.js, Express, PostgreSQL (Supabase), Passport.js, Google OAuth2, Vitest, ES Modules*

Street Keeper — *Final Year Project, University of Portsmouth*

- Designing and building a full-stack GPS analytics application that connects to Strava, helps users get into running, and turns street coverage into a game. Tracks which streets you have run, your completion percentage, and suggests where to run next. One local Portsmouth running group has indicated interest in adoption upon project completion.
- Built two distinct GPS analysis engines with documented trade-offs: V1 (Overpass API + optional Mapbox Map Matching, ~85% accuracy but with notable false positive rates) and V2 (CityStrides-style node proximity with pre-seeded OSM data, achieving 90% street-matching accuracy with zero recorded false positives or negatives).
- Architected a three-layer backend (Routes, Services, Data) using Node.js, Express 5, TypeScript (strict mode), and Prisma ORM with PostgreSQL. Designed a 12-table relational schema with indexed queries, cascade deletes, and a migration system for schema evolution.
- Implemented background job processing using pg-boss for asynchronous activity analysis, ensuring the frontend remains responsive during heavy computation.
- Integrated Strava OAuth 2.0 authentication with token refresh mechanism for long-term access. Implemented webhook subscriptions for automatic activity sync on new runs.
- Built the frontend in React 19, TypeScript, Vite, and Tailwind CSS 4. Used Context API for global state management, custom hooks for data fetching and geolocation, and a service layer for API communication.
- Implemented interactive map rendering with Leaflet: colour-coded street polylines (green/yellow/grey for complete/partial/unrun), clickable street cards with detailed stats, user location tracking, and map navigation controls.
- Designed a mobile-first responsive UI with WCAG 2.1 AA accessibility compliance: 44px minimum touch targets, keyboard navigation, screen reader compatibility, high contrast ratios, and semantic HTML structure.
- Implemented API rate limiting, JWT-based authentication, input validation at all layers, SQL injection prevention via Prisma, and CORS configuration.
- Wrote backend unit tests and API integration tests using Vitest. Documented frontend component tests and manual testing flows.
- Containerised with Docker. Environment-based configuration for development, staging, and production. Database migrations managed via Prisma.
- Integrated with external APIs: Strava (OAuth + webhooks), OpenStreetMap/Overpass (street geometry), and Mapbox (GPS trace snapping). Handled rate limits, error recovery, and fallback behaviour for each integration.

Stack: React 19, TypeScript (strict), Vite, Tailwind CSS 4, Leaflet, Node.js, Express 5, Prisma 7, PostgreSQL, pg-boss, Vitest, Docker, Strava API, Overpass API, Mapbox API, JWT, OAuth 2.0

Virtual Pet — Web Application

- Built a virtual pet web application featuring a custom-built lightweight JSON-based database layer for local data persistence (no external database dependency).
- Wrote a JavaScript-based conversational engine that parsed user text prompts and returned contextual, human-like responses using pattern matching and response templating.

Stack: JavaScript, HTML/CSS, custom JSON data layer

Hue Smash — Open-Source Web Game

- Developed and open-sourced a browser-based game designed to give Software Engineering classmates a collaborative coding experience.
- Structured the project to allow multiple contributors to work on independent features using Git-based workflows (branching, pull requests, code review).
- Acted as project lead: defined feature scope, reviewed contributions, resolved merge conflicts, and maintained code quality standards.

Stack: JavaScript, HTML5 Canvas/CSS, Git, open-source collaboration

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C#, Python, SQL (PostgreSQL, MySQL), PHP

Frontend: React, React Native, Vue, Angular, HTML5 Canvas, CSS3, Tailwind CSS

Backend: Node.js, Express.js, NestJS, .NET (C#)

Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM, Supabase

Cloud & DevOps: AWS (incl. CloudFormation, Lambda, Bedrock), Azure, Firebase, Render, Docker, CI/CD pipelines, Git/GitHub, pg-boss

Real-Time & Communication: WebSockets, Twilio (video, SMS), real-time chat systems, webhook integrations

Testing: Vitest (unit + integration, 230+ tests on Gloomhunt), component testing, manual test flow documentation, UAT tooling development

APIs & Payments: RESTful API design, OAuth 2.0 (Strava, Google), JWT, Passport.js, Strava API, Overpass/OSM API, Mapbox API, Stripe (subscriptions, webhooks, payment intents)

AI / ML: Amazon Bedrock (model training, knowledge bases, agent creation), Azure AI Language Studio (intent classification, entity extraction, NLU training), Cortex (Lilly internal LLM gateway), AWS Lambda (AI agent functions), chatbot framework development

Game Development: Custom engine development (vanilla JS + Canvas), spatial hash collision detection, seeded PRNG, deterministic game loops, object pooling, game AI (pathfinding, flocking, ability systems), anti-cheat architecture

Financial Systems: Double-entry ledger design, atomic transactions with row locks, settlement pipelines, reconciliation, idempotent operations

Hardware & Embedded: Fusion 360, SolidWorks, IMU sensor integration, embedded programming, sensor fusion concepts, PID control concepts

Design & Prototyping: Figma, UI/UX implementation from design specs
Methodologies: Agile, SAFe, Scrum, code review, pair programming, CI/CD workflows

CERTIFICATIONS

- SAFe Agile Practitioner - Eli Lilly and Company (2024)
- UK General Data Protection Regulation (GDPR) Certificate - Bay House School (2023)
- UK Safeguarding Training Certificate - TSAT / Thinking Schools Academy Trust (2023)

VOLUNTEER EXPERIENCE

Lilly Copilot Facilitator 2024
Eli Lilly and Company

- Facilitated an internal workshop on Microsoft Copilot usage, demonstrating practical workflows for productivity improvement across teams. Received positive feedback from attendees.

Python Programming Instructor 2022
Ngadi Foundation

- Taught Python programming fundamentals to teenagers and young adults. Led Q&A sessions, provided mentorship, and encouraged interest in IT careers.

Game Development Instructor (Remote) 2020
Inspire Bootcamp

- Volunteered during COVID-19 lockdown to teach game development concepts to 50+ children worldwide. Delivered interactive lessons using visual programming tools and supported students through project-based learning.

INTERESTS

Running, DIY engineering projects, candy making, UAV and rocketry experimentation, game development, open-source contribution, teaching and mentorship.